

Investigating the Impact of Climate Change on Landslide Triggering Rainfall Events in the French Alpine Massifs

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In the context of climate change and the induced intensification of extreme weather events, rainfall episodes exceeding landslide-triggering thresholds may become more frequent. This study aims to characterize the evolution of such events in the French Alpine massifs under various climate change scenarios.

The approach is based on a cross-analysis of observed meteorological data (SAFRAN-S2M reanalysis, 1958-2023) and simulated data (ADAMONT dataset from the DRIAS-Climat portal), which includes 20 GCM/RCM couples over the period 2006-2100. Three greenhouse gas emission scenarios are analysed (RCP2.6, RCP4.5, and RCP8.5). The analysis combines two complementary methodologies: (i) a narrative-based approach focusing on selected climate models illustrating contrasting trajectories (wetter, drier, rainier, etc.), and (ii) a statistical analysis of aggregated ADAMONT ensemble data to identify predominant trends across models. This dual approach aims both to account for inter-model variability and to provide a synthetic overview of potential impacts of climate change on gravitational hazards in the French Alps.

Our results highlight contrasted evolutions depending on the massif, altitude, season and scenario. While the northern and western Alpine massifs appear marginally impacted by mean climate-induced changes in liquid precipitation likely to trigger landslides based on current regional empirical rainfall thresholds, the southern massifs exhibit a trend toward increased frequency, duration and intensity of landslide triggering rainfall events. Moreover, higher greenhouse gas concentrations are associated with increased variability in extreme events across the entire French Alps. Our findings provide a scientific basis for refining risk assessment frameworks and guiding adaptation strategies in mountainous regions.

Mots-Clés: Climate change, landslides, Alpine massifs, precipitation, climate models, SAFRAN S2M reanalysis, emission scenarios, DRIAS Climat

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